Lecture 12

Objective priors: Part 1

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(Edited from the lecture notes by Brian J. Reich, NC State)

Informative versus non-informative priors

- In some cases informative priors are available.
- Potential sources include: literature reviews; pilot studies; expert opinions; etc.
- **Prior elicitation** is the process of converting expert opinions to prior distribution.
- For example, an expert might not comprehend an inverse-gamma PDF, but if they give you a point estimate and a spread, you can back calculate a and b.
- There are tools for this, such as `https://jeremy-oakley.shinyapps.io/SHELF-single/`

Informative versus non-informative priors

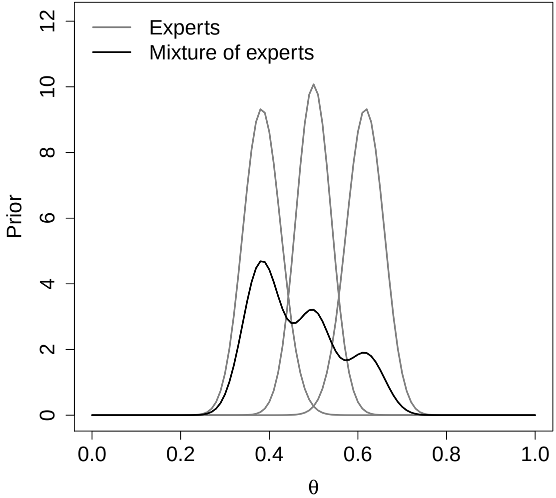
- Strong priors for the parameters of interest can be hard to defend.
- Strong priors for nuisance parameters are more common.
- For example, when you are doing a Bayesian t-test to study the mean μ, you might use an informative prior for the nuisance parameter σ^2.
- Anytime informative priors are used, you should conduct a **sensitivity analysis**.
- That is, compare the posterior for several priors.

Prior elicitation

- Prior information can come from a variety of sources.
- Say that source j (e.g., journal article, expert, pilot study) suggests prior $\pi_j(\theta)$.
- A **mixture of experts** prior combines the J sources into a single prior.
- Say source j is given weight $w_j > 0$ with $\sum_{j=1}^{J} w_j = 1$.
- The mixture of experts prior is

$$\pi(\theta) = \sum_{j=1}^{J} w_j \pi_j(\theta).$$

Mixture of experts prior ($w_j \in \{0.2, 0.3, 0.5\}$)



Mixture of experts prior

- The posterior distribution under the mixture of experts prior is

$$
\begin{aligned}
p(\theta|\mathbf{Y}) &\propto f(\mathbf{Y}|\theta) \times \pi(\theta) \\
&\propto \sum_{j=1}^{J} w_j f(\mathbf{Y}|\theta)\pi_j(\theta) \\
&\propto \sum_{j=1}^{J} Q_j p_j(\theta|\mathbf{Y}).
\end{aligned}
$$

- Here $p_j(\theta|\mathbf{Y}) \propto f(\mathbf{Y}|\theta)\pi_j(\theta)$ is the posterior under prior π_j.
- If each π_j is conjugate, the p_j belong to the same family as that of π_j.
- The posterior is also a mixture distribution over the same parametric family as the prior, and thus the prior is conjugate.

Prior elicitation

- Another powerful idea is to elicit prior information using real-life scenarios.
- "Do you expect more events on hot days than cold days?"
- "Would you expect more events on a rainy day in August or a sunny day in December?"
- Later you can frame these questions using the parameters in your model and back out the prior that best matches the expert responses.

Informative versus non-informative priors

- In most cases prior information is not available and so non-informative priors are used.
- Other names: “vague”, “weak”, “flat”, “diffuse”, etc.
- These all refer to priors with large variance.
- Examples: $\theta \sim \text{Uniform}(0, 1)$ or $\mu \sim \text{Normal}(0, 1000^2)$.
- Non-informative priors can be conjugate or not conjugate.
- The idea is that the likelihood overwhelms the prior.
- You should verify this with a sensitivity analysis.

Improper priors

- Extreme case: $\mu \sim \text{Normal}(0, \tau^2)$ and we set $\tau = \infty$.
- A "prior" that doesn't integrate to one is called an **improper** prior.
- Example: $\pi(\mu) = 1$ for all $\mu \in \mathbb{R}$.
- It's OK to use an improper prior so long as you verify that the posterior integrates to one.
- Subjective Bayesian interpretation is tricky.

Improper prior for the model Normal($\mu, 1$)

- Observations: $Y_1, \ldots, Y_n \overset{IID}{\sim}$ Normal($\mu, 1$). Let $\bar{Y} = (Y_1 + \ldots + Y_n)/n$.
- Likelihood: $f(Y_1, \ldots, Y_n|\mu) \propto \exp\left[-\frac{1}{2}\sum_{i=1}^{n}(Y_i - \mu)^2\right] \propto \exp\left[-\frac{n}{2}(\mu - \bar{Y})^2\right]$.
- Prior: $\pi(\mu) = 1$ for all $\mu \in \mathbb{R}$.
- Posterior: $p(\mu|Y_1, \ldots, Y_n) \propto f(Y_1, \ldots, Y_n|\mu)\pi(\mu) \propto \exp\left[-\frac{n}{2}(\mu - \bar{Y})^2\right]$
- Thus, $\mu|Y_1, \ldots, Y_n \sim$ Normal($\bar{Y}, \frac{1}{n}$).
- Despite the prior being improper, posterior is proper.

Subjective versus objective priors

- A subjective Bayesian picks a prior that corresponds to their current state of knowledge before collecting data.
- Of course, if the readers do not share this prior, then they might not accept the analysis, and so non-informative priors and a sensitivity analysis are common.
- An **objective analysis** is one that requires no subjective decisions by the analyst.
- Subjective decisions include picking the likelihood, treatment of outliers, transformations, ... and prior specification.
- A completely objective analysis may be feasible in tightly controlled experiments, but is impossible in many analyses.

Objective Bayes

- An objective Bayesian attempts to replace the subjective choice of prior with an algorithm that determines the prior.
- There are many approaches: Jeffreys, reference, probability matching, maximum entropy, empirical Bayes, penalized complexity, etc.
- Jeffreys priors are the most common in the literature.
- Many of these priors are **improper** and so you have to check that the posterior is proper.

Objective Bayes

- One subjective decision is the parameterization to use.
- For example, should we select $\sigma \sim \text{Uniform}(0, 10)$ or $\sigma^2 \sim \text{Uniform}(0, 100)$?
- These are quite different; $\sigma \sim \text{Uniform} \rightarrow p(\sigma^2) \propto 1/\sigma$.
- Any objective prior must give the same results, e.g., posterior mean for σ, for any parameterization.
- The Jeffreys prior is invariant to transformations.

Jeffreys prior

- The Jeffreys prior for parameter θ is $p(\theta) \propto \sqrt{I(\theta)}$.
- The Fisher information is $I(\theta) = -\mathrm{E}_{Y|\theta}\left[\frac{d^2}{d\theta^2}\log p(Y|\theta)\right]$.
- Once you have specified the likelihood, the Jeffreys prior is determined with no additional input.
- Therefore you do not have to make a subjective decision about the prior (other than "subjective" decision to use a Jeffreys prior).

Examples of Jeffreys priors

Likelihood	Jeffreys prior
$Y \sim \text{Binomial}(n, \theta)$	$\theta \sim \text{Beta}(1/2, 1/2)$
$Y \sim \text{N}(\mu, 1)$	$p(\mu) \propto 1$
$Y \sim \text{N}(0, \sigma^2)$	$p(\sigma) \propto 1/\sigma$

Jeffreys priors for the model $Y \sim \text{Binomial}(n, \theta)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = \log \binom{n}{x} + Y\log(\theta) + (n - Y)\log(1 - \theta).$$

- The first derivative is

$$\ell'(\theta) = \frac{Y}{\theta} - \frac{n - Y}{1 - \theta}.$$

- The second derivative is

$$\ell''(\theta) = -\frac{Y}{\theta^2} - \frac{n - Y}{(1 - \theta)^2}.$$

Jeffreys priors for the model $Y \sim \text{Binomial}(n, \theta)$

- Recalling that $E(Y) = n\theta$,

$$\begin{aligned} I(\theta) = -E(\ell''(\theta)) &= \frac{n\theta}{\theta^2} + \frac{n - n\theta}{(1-\theta)^2} \\ &= \frac{n}{\theta} + \frac{n}{(1-\theta)} \\ &= \frac{n}{\theta(1-\theta)}. \end{aligned}$$

- Finally, the prior is $\pi(\theta) \propto I(\theta)^{1/2} \propto \theta^{-1/2}(1-\theta)^{-1/2} = \theta^{1/2-1}(1-\theta)^{1/2-1}$.

- This the kernel of a Beta(1/2,1/2) PDF.

Jeffreys priors for the model $Y \sim \text{Normal}(\mu, 1)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2}\log(2\pi) - \frac{1}{2}(Y-\mu)^2.$$

- The first derivative is

$$\ell'(\theta) = Y - \mu.$$

- The second derivative is

$$\ell''(\theta) = -1.$$

- Therefore,

$$I(\theta) = -E(\ell''(\theta)) = 1.$$

- Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} = 1.$$

Jeffreys priors for σ for the model $Y \sim \text{Normal}(0, \sigma^2)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2}\log(2\pi) - \log(\sigma) - \frac{1}{2}\frac{Y^2}{\sigma^2}.$$

- The first derivative is

$$\ell'(\theta) = -\frac{1}{\sigma} + \frac{Y^2}{\sigma^3}.$$

- The second derivative is

$$\ell''(\theta) = \frac{1}{\sigma^2} - 3\frac{Y^2}{\sigma^4}.$$

- Therefore,

$$I(\theta) = -E(\ell''(\theta)) = -\frac{1}{\sigma^2} + 3\frac{\sigma^2}{\sigma^4} = \frac{2}{\sigma^2}.$$

- Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} \propto \frac{1}{\sigma}.$$

Thank you!